

Terafoods (TF) POS — Core Design System

Target & Context: Indoor Bangladeshi restaurants (Tier 2/3 small dine-in). Shared, budget Android devices (Helio G35/Snapdragon 4, 3–4GB RAM, 720p screens). Requires one-handed operation under indoor lighting.

Locale & Code: Bilingual (EN/BN). Source of truth: `tf_design_system.dart` + `app_theme.dart`.

1. **Core Principles Restraint:** Accent color ( #B8501F) covers $\leq 5\%$ of any screen (CTAs, focus rings). 95% remains warm near-monochrome. **Dominance Rule:** Maximum one full-saturation accent element per visible viewport. Transitive selected items must drop to a soft surface to keep the primary CTA dominant. **Glanceability:** Operators scan rather than read; every screen must answer its primary question in under 2 seconds. **Performance Constraint:** Visual decisions must respect the target low-end hardware. Zero-jank delivery over decorative animation.

Core Palette

`paper (#FAF8F5)`: Warm off-white scaffold background. Slightly yellow-warm — aged receipt paper, not clinical white.

`surface (#FFFEF9)`: Cards, sheets, fields, inputs. Warm breath — reads as white in practice.

`surfaceSunk (#F2EFE9)`: Search field fills, wells, depressed areas. Warm linen, not concrete.

`ink (#1A1410)` / `inkSoft (#2E2420)`: Primary text/headings, secondary dark. Espresso undertone — warm dark brown, not blue-black.

`muted (#6B6560)` / `mutedSoft (#9C9590)`: Secondary text/helpers/eyebrows, placeholders. Warm taupe, not cool grey.

`line (#E8E3DC)` / `lineStrong (#D4CEC6)`: Default 0.5 px borders, card outlines on paper. Warm cream and sand — not cool grey.

Accent (Burnt Orange) — Warm, Food-Coded, Distinctive

Rule: Accent never signals status or metrics. It is reserved purely for primary interactive states.

accent (#B8501F): Primary buttons, FAB, focused states, primary CTAs.

accentDeep (#8F3D14): Pressed states of accent elements.

accentSoft (#F5D9C8): Selected-state backgrounds (pairs with ink text on top).

accentWash (#FBEDE3): Subtle accent backgrounds, hovers.

accentInk (#FFFFFF): Text and icons on full-strength accent surfaces.

Functional Signals — Functional Only, Never Decorative

success (#15803D) / successSoft (#DCFCE7): Confirmed, healthy states, positive deltas only.

warning (#B45309) / warningSoft (#FEF3C7): Low stock, caution, capacity warnings.

danger (#7F1D1D) / dangerSoft (#FEE2E2): Out of stock, critical alerts, negative deltas.

urgent (#9A3412) / urgentSoft (#FFEDD5): Variance alerts, expedite signals.

3. Critical Color Logic

Deltas: Negative deltas use danger. Positive deltas use success. Accent is never used for data metrics.

Inventory & Stock States:

Low stock: Implements warning color tokens.

Out of stock (Alert State): Implements danger color tokens.

Intentionally empty SKU (Zero stock, no alert): Renders in neutral muted text on surface. Do not apply danger or warning treatments here.

Soft Surfaces: All soft signal surfaces (successSoft, warningSoft, dangerSoft, urgentSoft, accentSoft) pair exclusively with ink or their respective full-strength signal color. Never pair them with muted.

4. Typography & Hierarchy

Typography Tokens

Families: English → Inter, Bangla → Hind Siliguri.

Weights: Only w400 (Body, captions) and w500 (Headings, UI labels). w600+ is forbidden to protect low-res Bangla legibility.

Tracking: tight (-0.02em for hero scales), normal (0), wide (0.08em for uppercase eyebrows).

The Four Ladders

Ladder	Token	Size	Weight	Line Height	Primary Use Case
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Display					
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displayHero					
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displayLarge					
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44					
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36					
----	--	--	--	--	--

w500					
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w500					
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1.05					
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1.10					
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Single largest metric/total per screen.					
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Title					
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titleLarge					
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titleMedium					
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18					
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16					
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w500					
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w500					
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1.30					
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1.35					
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Card titles, alert/list headers. Pair with ink.					
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Body					
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bodyLarge					
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bodyMedium					
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bodySmall					
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15					
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14

12

w400

w400

w400

1.50

1.45

1.40

Readable text, descriptions, metadata captions.

Numeric (Tabular)

numericHero

numericMedium

44

18

w500

w500

Tabular Use for all pricing, quantities, currencies (₮).

Label

labelLarge

labelEyebrow

14

11

w500

w500

Normal

Wide

Button strings; uppercase section identifiers.

Bilingual Rule: BN text runs 20–30% longer than EN and requires a +1 px line height offset.

Fixed-width text wrappers are prohibited. Raw Text() widgets are banned; use TfText().

5. Spacing, Shapes, Layout & Shadows

Spacing (PosSpacing)

sp1 (4 px) | sp2 (8 px) | sp3 (12 px) | sp4 (16 px) | sp5 (24 px) | sp6 (32 px)

Standard Padding: Card contents & screen margins default to sp4 (16 px). Card sibling gap defaults to sp3 (12 px).

Shapes (PosRadii) & Borders

xs (6 px): Badges, chips | sm (10 px): Inputs, small buttons | md (14 px): Cards, dialogs | pill (999 px): FAB, active selections.

Borders: 0.5 px uniform width using line or lineStrong. No nested outlines.

Layout & Fixed Grids

Floor Map Grid: 4 columns on 720p screens. Gaps use sp3.

KPI rows / Quick Actions: 3 columns for status pulses; 4 columns for dashboard KPI tiles/actions.

Container Caps: Mobile runs full viewport width minus sp4 margins. Tablet layouts are capped at a maximum width of 720 px centered.

Stacking Order (Bottom Up)

System Navigation Inset → 2. TfBottomNav → 3. TfShiftStrip (Dashboard) OR TfStickyCTA (Flows) → 4. TfFab (Anchored bottom-right with sp4 margin from elements below).

Shadows (PosShadows)

none: Default (relies on borders).

soft (0 1px 2px rgba(26,20,16,0.05)): Stat tiles.

glow (0 4px 12px rgba(26,20,16,0.06)): Raised items, FABs.

raised (0 8px 24px rgba(26,20,16,0.08)): Sticky bottom CTAs, sheets.

6. Iconography (TfIcon)

Source: Lucide library only. 2 px stroke outlines; fills and animated morphs are banned.

Sizing (TfIconSize): xs (14) | sm (16) | md (20, Default) | lg (24) | xl (32) | xxl (48).

Inheritance: Icons inherit parental text color natively. Use explicit signaling colors only for functional context (e.g., green checkmark for success, red cross for danger).

7. Performance & Implementation Code

All static components must be compiled as const.

Lists larger than 10 entries must utilize lazy-loaded `ListView.builder` with predefined fixed `itemExtent`.

Image delivery must run through `cached_network_image`. Raw `Image.network` is banned. Menu items crop to 200×150 thumbnail files natively.